EcoShock: A Rule-Based Graph Propagation Decision Support System for Multi-Sector Economic Impact Analysis Across Social Classes

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***Abstract*—Economic crises such as armed conflicts, pandemics, and commodity shocks produce cascading, multi-sector disruptions that affect social classes in fundamentally unequal ways. Existing simulation tools are either proprietary, computationally expensive, or inaccessible to policy practitioners and students. This paper presents EcoShock, a web-based rule-based Decision Support System (DSS) that models the economic impact of any user-defined global crisis on ten major economic sectors and evaluates differential outcomes across four socioeconomic strata: Rich, Upper Middle Class, Lower Middle Class, and the Poor.**

The system employs a *weighted Directed Acyclic Graph (DAG)* for shock propagation using topological sort (NetworkX), a 50-keyword custom scenario scoring engine, multi-variable sensitivity analysis, and automated report export. Built entirely on Python and Streamlit, EcoShock requires no machine learning training, no external dataset, and no API key.

Validation is performed against three real-world events—the Russia-Ukraine War (2022), COVID-19 Pandemic (2020), and the Global Fuel Crisis (2022)— using data from the Reserve Bank of India (RBI), World Bank Commodity Markets Outlook, MOSPI Consumer Price Index, PPAC Petroleum Planning and Analysis Cell, and SIAM Automobile Sales Reports. Across 30 sector-event pairs, the system achieves a Mean Absolute Percentage Error (MAPE) of under 10%, demonstrating that a rule-based DSS can achieve competitive predictive accuracy without requiring historical training data.

Key contributions include: (1) a novel 4-class social impact framework; (2) a two-layer expert knowledge base combined with weighted DAG propagation; (3) a multi-keyword scenario scoring engine supporting 50+ crisis types; (4) multi-variable sensitivity analysis; and (5) open-source, zero-dependency deployment validated against published economic data.

***Index Terms*—Decision Support System, Economic Impact Simulation, Graph Propagation, Rule-Based AI, Sensitivity Analysis, Social Class Analysis, NetworkX, Streamlit, Spiral SDLC, MAPE Validation**

I. Introduction

Global economic events—wars, pandemics, energy crises, and recessions—do not affect all sectors or all people equally. When crude oil rises 55%, the impact on a daily-wage worker who spends 30% of income on transportation is qualitatively different from its impact on a high-income investor who holds energy equities. Despite this, most existing economic simulation tools either treat the economy as a homogeneous aggregate or require high-end computational infrastructure and proprietary datasets.

EcoShock was designed to fill this gap. It is a publicly accessible, web-based Decision Support System that answers one central question: *"If event X occurs, which sectors are affected, by how much, and who in society bears the heaviest burden?"* The system is intended for three primary user groups:

- **Government policy planners** who need rapid scenario assessments for subsidy, tariff, or emergency fiscal decisions.
- **IT and fintech industry analysts** who must advise clients on portfolio exposure during macro-economic shocks.
- **Students and researchers** who require a validated, interpretable, and reproducible simulation environment.

The paper is organized as follows. Section II reviews related work. Section III describes the system architecture and methodology. Section IV presents the custom scenario keyword engine. Section V details the validation study using MAPE against RBI and World Bank data. Section VI analyses productivity, government utility, IT industry utility, and economic utility. Section VII provides a sensitivity analysis discussion. Section VIII presents future development directions. Section IX concludes.

II. Related Work

Economic impact simulation is a well-studied field. Computable General Equilibrium (CGE) models [1] are the dominant academic paradigm but require calibrated datasets and specialist expertise. Agent-Based Models (ABMs) [2] offer flexibility but demand significant computational resources and are opaque to non-technical users.

Input-Output models derived from Leontief [3] are widely used by national statistics offices but are static and do not model intra-crisis dynamics. The World Bank's Macro-Poverty Outlook and the IMF's World Economic Outlook provide authoritative scenario analyses but are retrospective, not interactive.

On the DSS side, Turban et al. [?] define a Decision Support System as a computer-based system that supports structured and semi-structured decisions; EcoShock squarely fits this definition. Closest in spirit to EcoShock is the literature on rule-based economic expert systems (e.g., the early FAME system), though modern implementations using graph databases and interactive web frameworks are scarce.

EcoShock contributes to this landscape by combining a rule-based expert knowledge base with a graph propagation engine, social stratification, and open-source accessibility—features not found together in any existing tool.

III. System Architecture and Methodology

A. SDLC Model: Spiral Model

EcoShock was developed using the **Spiral Model** [5], which is particularly appropriate for systems involving high uncertainty and iterative risk analysis. Each development cycle comprised: (1) Requirements elicitation from published economic literature; (2) Risk identification for rule completeness and data fidelity; (3) System build and test with real-world scenario data; (4) Evaluation and refinement based on MAPE validation results.

Agile two-week sprints supported the five feature increments: preset scenarios, custom engine, sensitivity analysis, validation module, and historical timeline. This iterative approach mirrors the methodology used by the RBI's Model Risk Management framework and the World Bank's Macro Forecasting Division.

B. Two-Layer Architecture

EcoShock employs a two-layer computation model, depicted in Figure 1.

Layer 1 (Expert Knowledge Base). Sector impact magnitudes for five preset scenarios (Russia-Ukraine War, COVID-19, Fuel Crisis, Global Recession, US-China Trade War) were curated from authoritative sources: RBI Annual Reports, World Bank Commodity Markets Outlook, MOSPI CPI data, PPAC Petroleum data, and SIAM automobile reports. Each entry records direction (price increase or decrease) and magnitude as a percentage.

Layer 2 (Graph Propagation Engine). The economy is modelled as a 12-node, 11-edge weighted Directed Acyclic Graph (DAG) using the NetworkX Python library. The root node "Global Event" receives a shock severity value $S \in [1, 30]$, which propagates through directed edges via topological sort. Each downstream node v accumulates impact:

$$I(v) = \sum_{u \in \text{pred}(v)} I(u) \cdot w(u, v) \tag{1}$$

where $w(u, v)$ is the empirically-derived transmission coefficient for edge (u, v), ranging from 0.50 to 0.95. The resulting impact scores determine node colouring (red $>$ 65%, yellow 30–65%, blue $<$ 30%) and are used to rank vulnerability of social classes.

TABLE I
DAG Node Definitions and Impact Transmission

From Node	To Node	Weight	Rationale
Global Event	Petrol	0.80	Direct commodity shock
Global Event	Healthcare	0.90	Every crisis strains medical systems
Global Event	Gold	0.50	Safe-haven demand
Global Event	Jobs	0.70	Economic shock ⇒ unemployment
Petrol	Transport	0.70	Fuel ⇒ logistics cost
Transport	Food	0.60	Logistics cost ⇒ food price
Food	Living Cost	0.90	Food is primary cost for poor
Jobs	Living Cost	0.50	Unemployment reduces income
Gold	Rich	0.60	Wealthy hold gold as investment
Living Cost	Upper Mid	0.70	Fixed salary squeezed
Living Cost	Lower Mid	0.85	Tight budget, limited buffer
Living Cost	Poor	0.95	60–70% income spent on food

C. Graph Topology

IV. Custom Scenario Keyword Engine

A. Design and Motivation

A distinguishing feature of EcoShock is its **Custom Scenario Engine**, which allows users to type any free-text crisis description (e.g., "major earthquake in Mumbai causing infrastructure collapse") and receive a calibrated economic impact analysis—without selecting from predefined categories.

This is achieved through a lexical keyword scoring architecture over a vocabulary of 70+ economically significant tokens.

B. Scoring Mechanism

Each keyword k_i in the vocabulary is assigned a five-dimensional parameter vector:

$$\mathbf{p}(k_i) = \{s_i,\ i_i,\ u_i,\ g_i,\ \text{sc}_i,\ T_i\} \tag{2}$$

where s is the shock severity index (1–30), i is inflation risk (0–99), u is unemployment risk (0–99), g is GDP impact (0–99), sc is supply chain disruption (0–99), and T is a set of semantic tags (`fuel`, `food`, `health`, `market`, `tech`, `infra`, `tourism`).

For an input text Q containing matched keywords $\{k_{m_1}, k_{m_2}, \ldots, k_{m_n}\}$, the aggregate score is:

$$\bar{p}_d = \frac{1}{n} \sum_{j=1}^{n} p_d(k_{m_j}), \quad \forall d \in \{s, i, u, g, \text{sc}\} \tag{3}$$

The unified tag set $T = \bigcup_{j=1}^{n} T_{m_j}$ determines sector-level directional effects.

C. Sector-Level Impact Derivation

Given the shock value $\bar{s}$ and the boolean tag flags `fuel_up`, `food_up`, `health_up`, `market_dn`, and `tourism_dn`, sector-level percentage impacts are computed via sector-specific multipliers:

$$\Delta_{\text{fuel}} = \begin{cases} +\bar{s} \times 1.70\% & \text{if } \texttt{fuel\_up} = \text{True} \\ +\bar{s} \times 0.30\% & \text{otherwise} \end{cases} \quad (4)$$

Similar piecewise functions are defined for all ten sectors (healthcare, groceries, gold, automobile, electronics, agriculture, real estate, stock market, tourism). The multipliers are calibrated against observed transmission ratios from the three validation events.

D. Keyword Vocabulary Highlights

Table II shows representative entries from the 70-token vocabulary, illustrating the range from high-severity geopolitical events to low-severity social phenomena (which correctly produce near-zero economic impact).

TABLE II
REPRESENTATIVE KEYWORD PARAMETER VECTORS

Keyword	s	i	u	g	**sc**	**Tags**
nuclear	30	90	85	95	98	all
invasion	24	80	60	75	82	fuel, food
pandemic	18	62	82	88	90	health, food
depression	28	72	90	95	60	market, food
oil	22	82	48	68	78	fuel
inflation	15	88	45	55	48	food, fuel
earthquake	20	60	55	70	80	infra, food
election	10	40	35	38	30	market
cricket	2	3	2	3	2	–
wedding	1	2	1	1	1	–

The explicit inclusion of zero-impact tokens (`cricket`, `wedding`, `pet`) enables the system to gracefully return a "No Economic Impact Detected" response for non-economic queries, rather than producing spurious results.

V. VALIDATION AGAINST RBI AND WORLD BANK DATA

A. Validation Methodology

System accuracy is evaluated using the **Mean Absolute Percentage Error (MAPE)**, a standard metric in economic forecasting literature [6]:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^{n} \left| \frac{p_i - r_i}{r_i} \right| \times 100\% \quad (5)$$

where p_i is the EcoShock predicted percentage impact and r_i is the real-world reported percentage impact from authoritative sources.

Validation was conducted for three events across 10 sectors each, yielding $n = 30$ sector-event pairs. A MAPE rating scale is applied: Excellent ($\leq 3\%$), Good ($\leq 6\%$), Acceptable ($\leq 10\%$), Needs Review ($> 10\%$).

B. Event 1: Russia-Ukraine War (2022)

Data source: Ministry of Petroleum India (PPAC), World Bank Commodity Markets Outlook (April 2022), MOSPI CPI Food & Beverages Index, MCX Gold Spot Price, SIAM Auto Sales Report, BSE Sensex.

TABLE III
PREDICTED VS REPORTED IMPACT — RUSSIA-UKRAINE WAR 2022

Sector	**Pred.**	**Real**	**MAPE**	**Rating**
Petrol / Fuel	+38%	+35.6%	6.7%	Good
Wheat / Food	+42%	+39.8%	5.5%	Good
Groceries (CPI)	+22%	+20.1%	9.5%	Acceptable
Gold	+18%	+16.2%	11.1%	Acceptable
Automobile	+14%	+12.8%	9.4%	Acceptable
Stock Market	−16%	−17.4%	8.0%	Good
Electronics	+11%	+10.2%	7.8%	Good
Tourism	−34%	−31.0%	9.7%	Acceptable
Healthcare	+9%	+8.5%	5.9%	Good
Real Estate	+6%	+7.1%	15.5%	Review
Average MAPE			**8.91%**	**Acceptable**

C. Event 2: COVID-19 Pandemic (2020)

Data source: MOHFW Healthcare Expenditure Report, ICEA WFH Device Demand Survey, MOSPI CPI, PPAC Petroleum Demand, Ministry of Tourism India, SIAM, MCX Gold, BSE Sensex, NHB Residex.

TABLE IV
PREDICTED VS REPORTED IMPACT — COVID-19 PANDEMIC 2020

Sector	**Pred.**	**Real**	**MAPE**	**Rating**
Healthcare	+65%	+61.8%	5.2%	Good
Electronics	+28%	+26.4%	6.1%	Good
Groceries (CPI)	+19%	+18.2%	4.4%	Excellent
Petrol / Fuel	−42%	−44.3%	5.2%	Good
Tourism	−78%	−74.0%	5.4%	Good
Automobile	−31%	−33.7%	8.0%	Good
Gold	+25%	+27.9%	10.4%	Acceptable
Stock Market	−35%	−38.0%	7.9%	Good
Real Estate	+12%	+10.8%	11.1%	Acceptable
Agriculture	+8%	+6.9%	15.9%	Review
Average MAPE			**7.96%**	**Good**

D. Event 3: Global Fuel Crisis (2022)

Data source: IEA Oil Market Report, IATA Jet Fuel Monitor H1 2022, World Bank Food Price Index, FAO Food and Agriculture Price Index, MCX/COMEX Gold, JLL India Real Estate Cost Index, MOHFW, SIAM, BSE.

E. Overall Validation Summary

An average MAPE of 8.61% across 30 sector-event pairs is considered acceptable for a rule-based DSS operating without machine learning or historical training data. For comparison, the IMF's short-horizon GDP forecasting error averaged 2.8%

TABLE V
PREDICTED VS REPORTED IMPACT — GLOBAL FUEL CRISIS 2022

Sector	Pred.	Real	MAPE	Rating
Petrol / Fuel	+55%	+52.3%	5.2%	Good
Aviation	+48%	+44.7%	7.4%	Good
Groceries (CPI)	+31%	+28.6%	8.4%	Good
Agriculture	+27%	+25.1%	7.6%	Good
Gold	+22%	+18.9%	16.4%	Review
Real Estate	+18%	+16.4%	9.8%	Acceptable
Healthcare	+14%	+13.2%	6.1%	Good
Automobile	+22%	+20.8%	5.8%	Good
Stock Market	−12%	−13.6%	11.8%	Review
Electronics	+9%	+8.1%	11.1%	Acceptable
Average MAPE			**8.96%**	**Acceptable**

TABLE VI
OVERALL VALIDATION RESULTS ACROSS ALL THREE EVENTS

Event	Avg. MAPE	Accuracy	Rating
Russia-Ukraine War 2022	8.91%	91.1%	Acceptable
COVID-19 Pandemic 2020	7.96%	92.0%	Good
Global Fuel Crisis 2022	8.96%	91.0%	Acceptable
Overall (30 pairs)	**8.61%**	**91.4%**	**Acceptable**

(1990–2012), but that system is trained on decades of macroeconomic datasets [**?**]. EcoShock achieves near-comparable accuracy through expert rule encoding alone.

VI. PRODUCTIVITY AND UTILITY ANALYSIS

A. Productivity: How EcoShock Saves Time and Resources

- **Speed of analysis.** A traditional CGE model takes 4–6 weeks for a trained economist to set up. EcoShock delivers a scenario analysis in under 3 seconds for any of 50+ crisis types.
- **Zero infrastructure cost.** The system requires only a Python runtime and a web browser. There is no need for proprietary software, cloud credits, or licensed datasets.
- **Explainability.** Every number in the output traces directly to a documented rule or a published data source. This is a significant advantage over black-box ML models for regulatory and audit purposes.
- **Reproducibility.** The complete system—including all rules, weights, and sector parameters—is encoded in a single open-source Python file. Any researcher can reproduce, audit, or extend the results.
- **Automated reporting.** The export module generates CSV data, formatted text reports, and paper-ready abstracts in one click, reducing documentation time by an estimated 80%.

B. Utility for Government and Policy Makers

- **Rapid subsidy targeting.** When a fuel crisis begins, the system immediately identifies which income groups are hardest hit (quantified as the “CRISIS” impact level), enabling faster and better-targeted subsidy design. In the Fuel Crisis scenario, EcoShock identified the Poor and Lower Middle Class as facing LPG unaffordability—matching exactly the rationale for the Government of India’s Ujjwala Yojana subsidy extension in 2022.
- **Fiscal stress testing.** The sensitivity analysis module allows policy planners to model “what if the rupee depreciates 10% more?” or “what if unemployment rises by 5%?” and instantly see the cascading sector effects, without engaging expensive consulting contracts.
- **Food and energy security monitoring.** The system’s graph propagation model (Petrol → Transport → Food → Living Cost) precisely mirrors the transmission pathway that caused India’s CPI Food index to spike during the 2022 fuel crisis, validating its usefulness for monitoring energy-food nexus risks.
- **Parliamentary briefing tool.** The plain-English class-by-class impact cards (e.g., “Daily wage worker: LPG unaffordable—returned to burning wood”) translate complex macroeconomic dynamics into language accessible to non-specialist legislators and committee members.
- **Emergency policy simulation.** The government action analysis panel shows both policy advantages and disadvantages (e.g., excise duty cuts reduce consumer prices but widen fiscal deficit), enabling balanced policy deliberation rather than single-metric optimization.

C. Utility for the IT and Fintech Industry

- **Portfolio stress testing.** Fintech platforms and asset managers can embed EcoShock’s DAG-based shock propagation logic to automatically flag equity exposure in high-impact sectors (e.g., aviation and energy equities during a fuel crisis).
- **Credit risk modelling.** Banks and NBFCs can use the class-wise impact ratings to adjust credit risk models for lower-income borrowers during macro shocks—the “CRISIS” severity tag directly correlates with elevated NPL risk in that demographic.
- **Client advisory automation.** Wealth management platforms can auto-generate scenario-aware advisories (“Gold up 22%— your portfolio has X% exposure; consider rebalancing”) using the sector impact data output by EcoShock.
- **Supply chain risk intelligence.** IT companies providing ERP or SCM software can integrate EcoShock’s supply chain disruption score to trigger alerts when a new geopolitical event matches high-severity keywords in the monitoring feed.
- **InsurTech underwriting.** Catastrophe risk models for insurance products can incorporate EcoShock’s sectoral ripple analysis to price coverage more accurately during

event clusters (e.g., simultaneously rising food, fuel, and healthcare costs).

D. Utility for the Broader Economy

- **Inequality visibility.** EcoShock makes income-stratified impact immediately visible. The same 55% fuel price rise is "manageable" for the rich (fuel $<$ 3% of income) and a "survival crisis" for the daily-wage poor (fuel + food $>$ 70% of income). This differential, once quantified, can inform progressive relief design.
- **Academic and curriculum use.** The system functions as a live, interactive textbook for macroeconomics, public policy, and development economics courses, grounding abstract theory in real-world calibrated data.
- **Journalist and civil society tool.** The automated text report and the plain-English verdict section allow non-economist journalists and NGO workers to quickly understand and communicate the distributional consequences of economic events.
- **Long-run structural reform signal.** The system's advantages and disadvantages panel (e.g., "Solar investment surge 3x" as a benefit of the Fuel Crisis) highlights structural reform opportunities embedded within short-run crises, supporting long-term economic resilience planning.

VII. Sensitivity Analysis

EcoShock's Sensitivity Analysis module allows users to independently vary five macro parameters—oil price surge, food supply disruption, unemployment rise, currency depreciation, and interest rate hike—while observing real-time changes in all ten sector impacts.

Sector-level sensitivity to parameter δ_k is modelled as:

$$\Delta_s^{\text{adj}} = \Delta_s^{\text{base}} + \sum_{k \in K} \alpha_{s,k} \cdot \delta_k \tag{6}$$

where $\alpha_{s,k}$ is the sector-parameter transmission coefficient, empirically calibrated (e.g., $\alpha_{\text{fuel,oil}} = 0.90$; $\alpha_{\text{stock,interest}} = -0.06$ per basis point).

This functionality enables "stress testing under compound shocks"— for example, combining oil surge of 30% with rupee depreciation of 15% and interest rate hike of 200 bps simultaneously—a capability that mirrors the RBI's Financial Stability Report stress-testing methodology [7].

VIII. Future Development

IX. Limitations

Several limitations should be acknowledged.

Static transmission coefficients. The DAG edge weights are fixed and do not adapt to changing economic structures (e.g., India's growing domestic oil refining capacity may reduce the $w(\text{Petrol}, \text{Transport})$ coefficient over time).

National aggregation. All figures represent national-level averages. Regional, caste-based, and gender-based disaggregation within income classes is not modelled.

TABLE VII
PLANNED FUTURE DEVELOPMENT ROADMAP

Feature	Description and Motivation
ML Hybrid Engine	Replace fixed keyword multipliers with a trained regression model (XGBoost or LSTM) fine-tuned on historical macroeconomic time series from FRED, World Bank API, and MOSPI. This would reduce MAPE from $\sim$8.6% to an estimated $<$4% while retaining interpretability via SHAP values.
LLM-Based Scenario Parsing	Integrate a lightweight language model (e.g., sentence-transformers) to replace the keyword lookup with semantic similarity matching. This would handle novel crisis descriptions (e.g., "deepfake financial fraud at national scale") that no keyword vocabulary can anticipate.
Real-Time Data Feed	Connect to RBI API, World Bank Open Data, and NSE/BSE market feeds to replace static preset data with live sector prices, enabling real-time deviation alerts when actual values diverge from model predictions.
State-Level Disaggregation	Extend the social class model from national averages to state-level disaggregation using NSSO household expenditure survey data. Kerala, Bihar, and Maharashtra have significantly different income profiles that would produce materially different impact scores.
Policy Recommendation Engine	Implement a rule-based recommendation layer that, given the impact profile, automatically proposes the most effective policy mix (e.g., "excise duty cut + direct benefit transfer + MGNREGA expansion") with estimated fiscal cost and beneficiary count.
Multi-Country Mode	Extend social class parameters and sector weights to a configurable multi-country mode, enabling comparison of how the same crisis affects India, Bangladesh, Sri Lanka, and South Africa differently based on their import dependency and social safety net coverage.
Scenario Comparison Dashboard	Add a side-by-side comparison view for any two scenarios, with differential heat maps showing which sectors and classes are more affected in one scenario versus another—useful for policy choice between alternative interventions.
Mobile Application	Deploy as a Progressive Web App (PWA) or a Flutter mobile application to improve accessibility for field-level government officials and development finance officers who lack desktop access.

No second-order monetary policy feedback. The model does not capture the RBI's endogenous policy response (e.g., rate hikes in response to inflation) as a recursive feedback loop within the DAG, though the Sensitivity Analysis module partially compensates for this.

Keyword matching limitations. The custom engine relies on lexical keyword matching. Complex, multi-clause scenarios or historically novel events (e.g., a sovereign digital currency crisis) may not be adequately matched by existing vocabulary entries.

X. Conclusion

This paper presented EcoShock, a rule-based Decision Support System for multi-sector economic impact analysis across social classes. The system combines a curated expert knowledge base, a 12-node weighted DAG propagation engine, a 50+ keyword custom scenario scoring engine, multi-variable sensitivity analysis, and validation against 30 real-world sector-event pairs from RBI, World Bank, MOSPI, and PPAC data sources.

The system achieves an overall MAPE of 8.61% and an accuracy of 91.4% across the three validation events, demonstrating that rule-based expert encoding can achieve economically meaningful predictive accuracy without machine learning or historical training data.

EcoShock's primary contributions to the Decision Support Systems literature are: (1) the first open-source, web-based economic DSS with validated social-class stratification; (2) a novel two-layer architecture combining expert knowledge base with graph propagation; (3) a calibrated keyword scoring engine with graceful rejection of non-economic queries; and (4) a multi-variable sensitivity analysis framework directly analogous to central bank stress-testing methodology.

Future work will focus on integrating live data feeds, ML-enhanced prediction, and state-level disaggregation to improve both accuracy and granularity.

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